

RETAILSPHERE (PTY) LTD

Data Platform Programme

Metadata Strategy

Step 11 Deliverable — Metadata Strategy

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Author	Thozamile Mbalo – Data Solutions Architect
Reviewed by	Nomvula Mahlangu – Chief Data Officer
Target Platform/Tooling	dbt docs / Microsoft Fabric Warehouse
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1. Purpose

This document defines RetailSphere's approach to metadata: how data lineage is maintained, how a data dictionary is captured and kept current, and what documentation standard every column in the platform must meet. It satisfies FR-CCR-001 — every number in the Gold layer must be traceable back to its source transaction through Bronze and Silver.

Beyond the stated requirement, this document exists because undocumented systems become unmaintainable systems. A platform without a metadata discipline degrades the same way any system does when the people who understood it move on and nothing was written down — RetailSphere is designed from Step 11 onward to never reach that state.

2. Scope

MET-001 covers lineage, data dictionary maintenance, and column-level documentation standards across all three medallion layers (Bronze, Silver, Gold). It defines *what* must be documented and *to what standard* — it does not itself contain the documentation (that lives in dbt's **schema.yml** files and generated docs site, built during Step 14). This document is the standard those artifacts must meet, not a substitute for them.

3. Lineage (FR-CCR-001)

FR-CCR-001 requires that every number in the Gold layer be traceable back to its source transaction, within minutes, through Bronze and Silver.

Model-level lineage is not sufficient on its own. dbt automatically generates a model-level lineage graph from the SQL relationships between models — no manual effort required, this comes free with **dbt docs generate**. But model-level lineage only shows *which models* feed which — it does not by itself confirm *which column* a specific number trace to. A hidden intermediate layer between systems is a common real-world failure mode: an apparent replacement or migration of one layer can create false confidence that a dependency chain has been fully resolved, when in fact an undocumented layer underneath continues to carry the real dependency. Column-level lineage is what prevents this — knowing precisely that **FactFinance.debit_amount** traces to a specific source field, not just "the Finance domain traces to SAP ECC" in general terms, is what actually satisfies "traced within minutes" rather than "traced eventually, after investigation."

Documentation discipline: incremental, not retroactive. Column-level descriptions in **schema.yml** are not generated automatically — they must be written by whoever builds each model. This documentation is written **incrementally, as each model is built**, not as a batch exercise after Step 14's build phase completes. Retroactive documentation reliably misses detail that was obvious at build time and easily forgotten weeks later — the same failure mode that leaves any complex system dependent on the memory of the few people who built it.

Standard: Every dbt model's **schema.yml** must include a description for the model itself and for every column, at the time that model is built — not deferred, not batched, not left as a follow-up task.

4. Data Dictionary

The data dictionary is not a separate, manually maintained artifact — it is generated directly from the same **schema.yml** descriptions written incrementally during model build (Section 3). This avoids the common failure mode of a maintained wiki or spreadsheet drifting out of sync with what the pipeline actually does.

Primary users — engineers: dbt's auto-generated docs site (**dbt docs generate**) is the technical source of truth. It is not directly editable; changes are made in **schema.yml** and the site is regenerated. This is the expected tool for anyone building or extending a model.

Secondary users — business stakeholders: Johan, Priya, Rethabile, and other domain owners are not expected to use dbt's technical docs site directly. Column and measure definitions are instead surfaced through Power BI's own description fields, populated from the same underlying **schema.yml** definitions — giving business users plain-language definitions in the tool they already work in daily (per DID-001's Consume layer), rather than introducing a separate data catalog product RetailSphere has not scoped or budgeted for.

A dedicated enterprise data catalog tool (e.g. Microsoft Purview, Collibra) may become worth evaluating as the platform scales beyond its current six domains — flagged as a future consideration, not a current requirement.

5. Column-Level Documentation Standard

Every column across Bronze, Silver, and Gold must carry the following, written into **schema.yml** at build time (Section 3's incremental discipline):

Field	Source	Notes
Data type	PDD-001	e.g. MONEY, VARCHAR (20), BIGINT
Business Description	DM-001	Plain-language meaning, not just the column name restated
Source system and field	STM-001	Traces to the actual origin system/table/field
Transformation logic	STM-001	Any calculation, mapping, or derivation applied
Confidence rating	STM-001	Carried forward from design — see below
Sample value	New – this standard	Synthetic only, never real production data, regardless of project
POPIA sensitivity classification	New – this standard	See tiered scheme below

This consolidates what currently requires opening three separate documents (DM-001, STM-001, PDD-001) into a single, queryable reference per column.

Confidence rating, redefined for production use: STM-001's **CONFIRMED/ASSUMED** distinction is carried forward, but its meaning shifts once a pipeline is live. It no longer indicates whether the mapping *works* — a running, tested pipeline has already proven that. It indicates **how well the mapping was validated by a subject matter expert** at design time. A **CONFIRMED** field was SME-verified; an **ASSUMED** field that happens to work today is still an unconfirmed guess that could silently break if the source system changes in a way nobody explicitly agreed to. This distinction matters most for prioritizing which fields to re-validate first if a source system changes.

POPIA sensitivity classification (tiered):

Known limitation: this classification scheme operates at the single-column level, matching **schema.yml**'s structure. It cannot currently express risk that only emerges from a **combination** of columns viewed together — e.g. customer segment combined with contact history, which may constitute a higher re-identification or profiling risk than either column carries alone. This is a genuine gap, not yet solved by this standard, and is flagged here rather than papered over with an example the mechanism cannot actually support. Addressing this may require view-level or query-level sensitivity tagging, which is out of scope for MET-001 v1.0.

Tier	Meaning	Example
None	No personal or sensitive business data	date_key, store_key
Internal	Business-sensitive but not personal data; visible to staff, not for external sharing	contracted_rate, unit_cost
Sensitive – PII	Personal information; access should be role-restricted	email, phone_number, physical_address
Highly Sensitive	PII combined with financial or behavioural profiling; requires the strictest access control	Customer segment + contact history combined views

A binary flag cannot distinguish "visible to all staff" from "requires restricted access or must never be exported externally" — the tiered scheme exists specifically to support that distinction, which matters given ADR-002 and ADR-007's POPIA obligations are legal requirements, not preferences.

6. Governance and Ownership

Enforcement — automated, not memory-dependent. Rather than relying on individual engineers remembering to document their work, or a lead manually checking every change, documentation completeness is enforced as an **automated build gate**: a dbt build fails if any new or changed model or column lacks a description. This applies the same principle already established for data quality (Section 4's schema.yml-driven dbt tests) to documentation itself — the pipeline, not a person's memory, is what actually prevents drift at the point of change. This gate is tested and confirmed live during Step 12, before the first model is built in Step 14 — not introduced partway through the build. This ensures no column is ever built without documentation being enforced from the very first commit, avoiding a backlog of undocumented "legacy" columns that predate the gate's existence.

Ownership roles:

- **Lead Data Engineer** owns the documentation standard itself, handles genuine exceptions the automated gate can't judge (e.g. a column whose meaning is legitimately still evolving), and liaises directly with domain owners on business-definition accuracy. RetailSphere has no separate product manager role — this liaison responsibility sits with the lead engineer directly.
- **Domain owners** (Sipho, Zanele, Rethabile, Priya, Johan) are accountable for the *business accuracy* of definitions in their domain — confirming that what's documented actually reflects how the business understands the data, not just that a description field is technically filled in.
- Accountability is **shared, not exclusive to either side** — an "or" structure between engineering and domain ownership was deliberately rejected, since it creates space for each party to assume the other is responsible.

Review cadence: A quarterly documentation audit catches drift the automated gate cannot — cases where a description exists but has become stale or inaccurate relative to what the business or the pipeline actually does now. This audit is a scheduled check, not a substitute for the build-time gate; it exists specifically to catch decay in things that technically pass the automated check but have quietly gone wrong.

Accountability if FR-CCR-001's promise breaks: If a Gold layer number cannot be traced within minutes due to stale or missing documentation, this is treated as a **shared platform governance failure**, not attributed solely to engineering or solely to a domain owner. Consistent with the CDO review pattern established throughout this project (PDD-001, DID-001), this is exactly the class of finding a CDO review would surface directly — and the automated gate plus quarterly audit exist specifically to prevent it from reaching that point.

7. Next Steps

The completion of MET-001 marks the end of Step 11 — Metadata Strategy. The following items carry forward:

Immediate:

- Publish MET-001, update Notion step tracker and README Step 11 status.
- Run a CDO review pass (same format as PDD-001 and DID-001) before MET-001 is marked Approved.

Step 12 — Testing Strategy: Define the platform's testing approach — building directly on Section 3's dbt test gates (FR-CCR-007) and Section 6's automated documentation gate, extending them into a full test plan covering unit, integration, and reconciliation testing across all six domains.

Outstanding items requiring resolution before build (Steps 13+):

- **Automated documentation-completeness gate** (Section 6) — the dbt CI configuration that actually enforces this must be built and proven during Step 14, not just designed here.
- **Quarterly documentation audit** (Section 6) — schedule and format to be formally established once Step 14 build begins and there is real documentation to audit.
- **POPIA sensitivity classification** (Section 5) — must be applied per column during Step 14 build; tied to ADR-002 and ADR-007's outstanding sign-offs, since the classification scheme's real-world stakes depend on those approvals completing.
- **Enterprise data catalog evaluation** (Section 4) — deferred; revisit only if the platform scales meaningfully beyond its current six domains.
- **Carried forward from PDD-001 and DID-001:** ADR-002/003/006/007 sign-offs remain in progress; FactContactHistory attempt-sequencing validation against real Marketing Cloud behaviour is still unresolved.

8. Document Change Log

Version	Date	Author	Description
1.0 - Draft	16 September 2026	Thozamile Mbalo	Initial MET-001 draft. Defines lineage strategy (FR-CCR-001) built on dbt's native lineage graph with incremental documentation discipline; data dictionary approach using dbt docs for engineers and Power BI field descriptions for business stakeholders, avoiding a separate uncontracted data catalog tool; column-level documentation standard consolidating fields from DM-001, STM-001, and PDD-001 into one location, with a tiered POPIA sensitivity classification and a synthetic-data-only rule for sample values; governance model using an automated dbt build gate for documentation completeness (not manual enforcement), shared accountability between the Lead Data Engineer and domain owners, and a quarterly audit cadence to catch drift the automated gate cannot.
1.0 – Approved	16 September 2026	Reviewed by Nomvula Mahlangu, Chief Data Officer	Approved following review. Section 6 clarified: automated documentation gate confirmed tested and live from Step 12, before Step 14's first model is built — no backlog risk. Section 5: flagged a known limitation — the column-level POPIA sensitivity scheme cannot express risk arising from combinations of columns (e.g. segment + contact history); noted honestly as unsolved rather than removed or overstated.

— End of Document —